



NOVA
IMS
Information
Management
School

Data Mining I

S6

NOVA-IMS
Fernando Lucas Bação
fbacao@isegi.unl.pt
<http://www.isegi.unl.pt/fbacao>

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

1

NOVA
IMS
Information
Management
School

Clustering

Agenda

- Cluster analysis
- Variables to use
- Similarity criterion
- Clustering algorithms
 - A Priori Grouping
 - RFM Analysis

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

2

NOVA
IMS
Information
Management
School

Cluster Analysis

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

UNICIS

ASES

Schools

eduniversal

3

NOVA
IMS
Information
Management
School

Clustering

- **Cluster Analysis**
 - Cluster Analysis is a **basic conceptual activity of human beings**;
 - A **fundamental process**, common to many sciences, essential to the development of scientific theories;
 - The possibility of **reducing the infinite complexity of real** to sets of objects or similar phenomena, is one of the most powerful tools in the service of mankind.

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

4

NOVA
IMS
Information
Management
School

Clustering

- **Cluster Analysis**
 - Cluster analysis is a generic name for a variety of methods that are used to **group entities**;
 - Objective: **To form groups of objects that are similar to each other**;
 - From a data collection about a group of entities, seeks to organize them **in homogeneous groups**, assessing a "frame" of similarities/differences between units.

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

5

NOVA
IMS
Information
Management
School

Clustering

- **Cluster Analysis**

The diagram shows a 3D coordinate system with three clusters of points. The first cluster (pink points) is circled in red, and a callout box labeled 'Minimize intra-cluster distances' points to it. The second cluster (blue points) is circled in red, and a callout box labeled 'Maximize intra-cluster distances' points to it. The third cluster (orange points) is circled in red, and a callout box labeled 'Maximize intra-cluster distances' points to it.

Source: Tan, Steinbach, Kumar, Introduction to Data Mining 2004

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

6

NOVA
IMS
Information
Management
School

Clustering

- **Cluster Analysis**
 - Classification:
 - Starts out with a **pre-classified training set**, that is, the method has a set of data which contains not only the variables to use in classification but also the class to which each of the records belongs;
 - Attempts to develop a model capable of predicting how a new record will be classified.

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

7

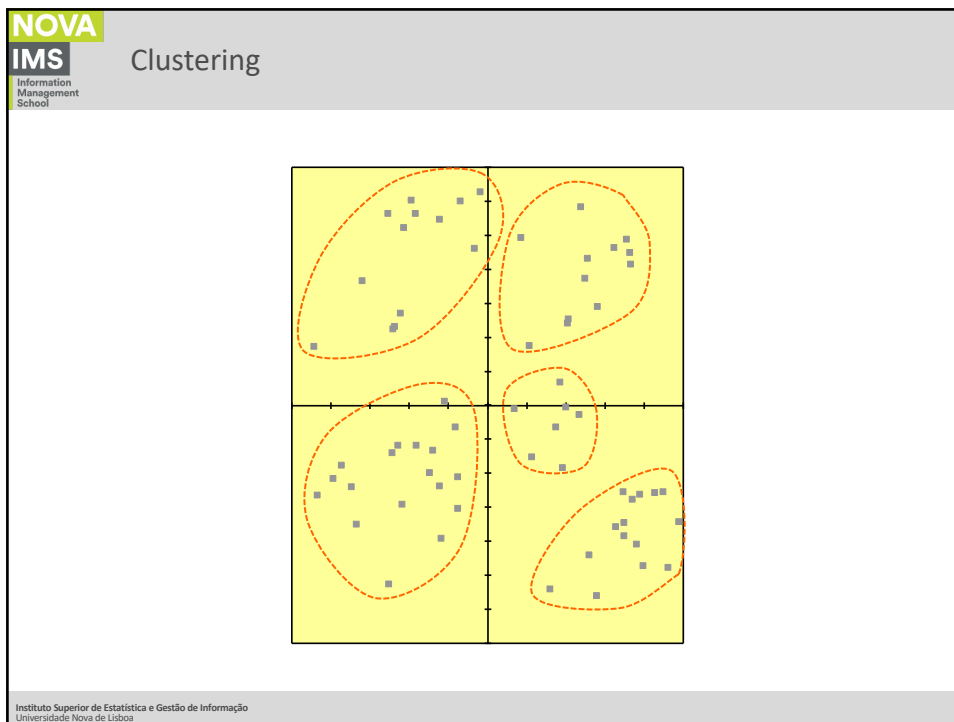
NOVA
IMS
Information
Management
School

Clustering

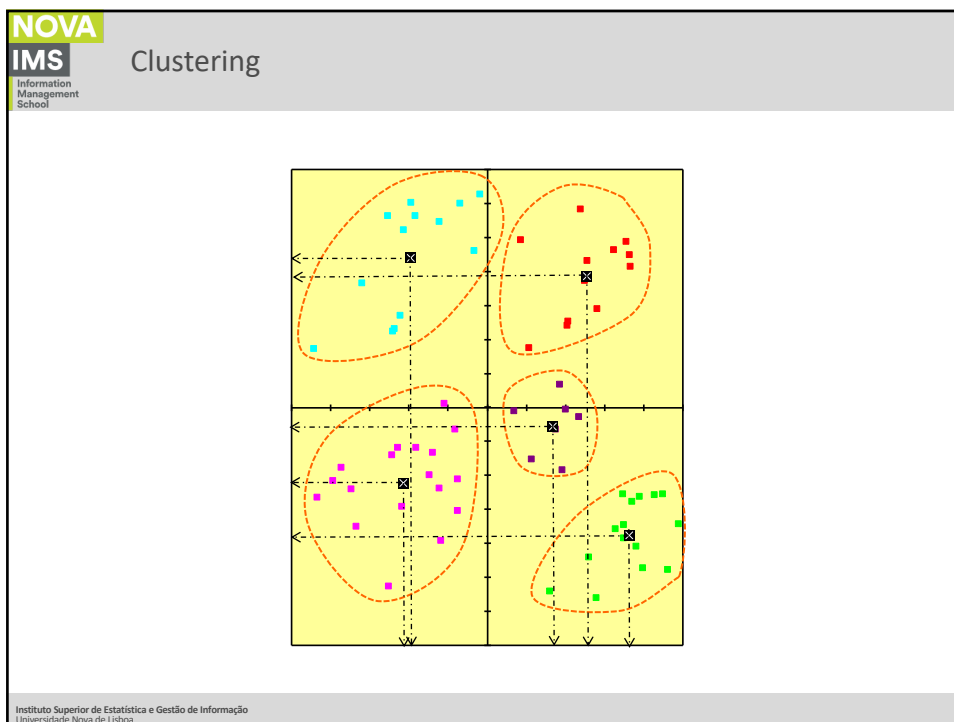
- **Cluster Analysis**
 - Clustering:
 - There is **no pre-classified data**;
 - We search for groups of records (clusters) that are similar to one another;
 - Underlying is the expectation that similar customers in terms of the variables used will behave in similar ways.

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

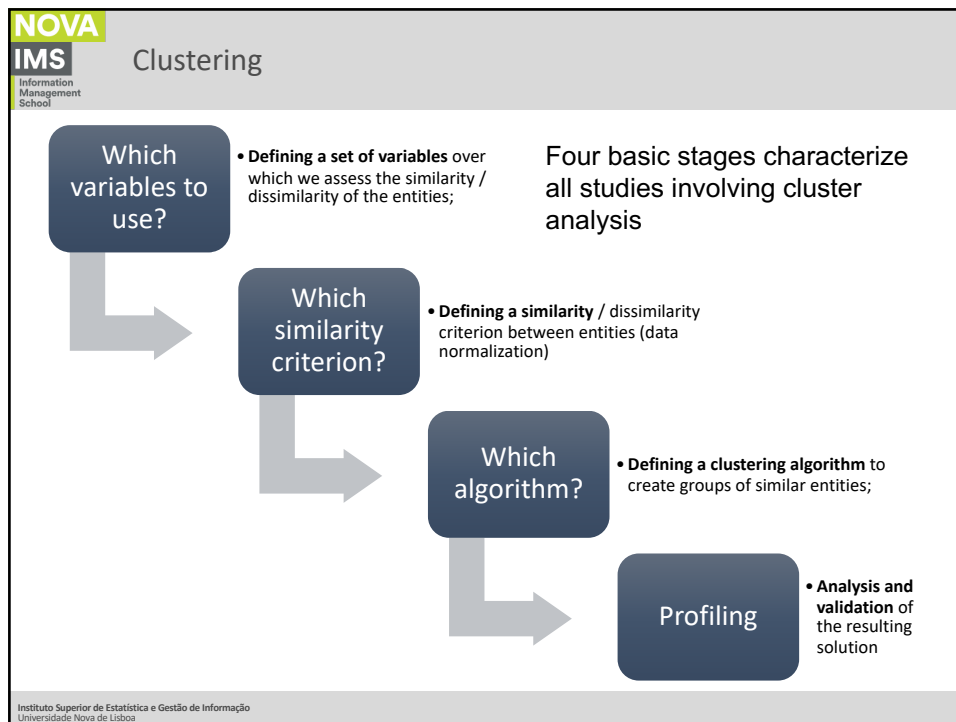
8



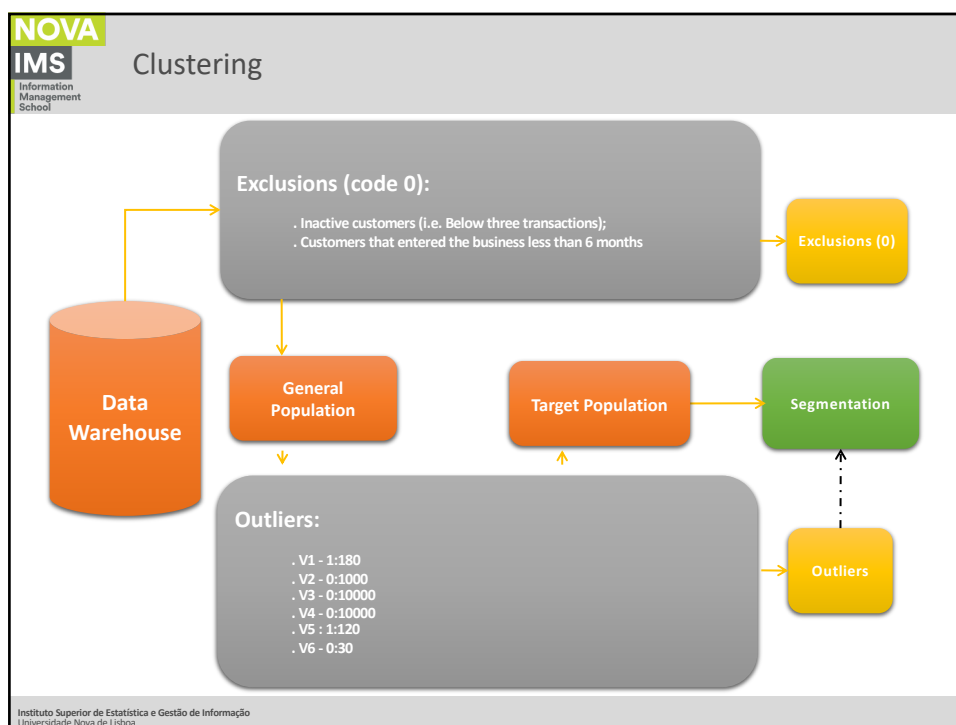
9



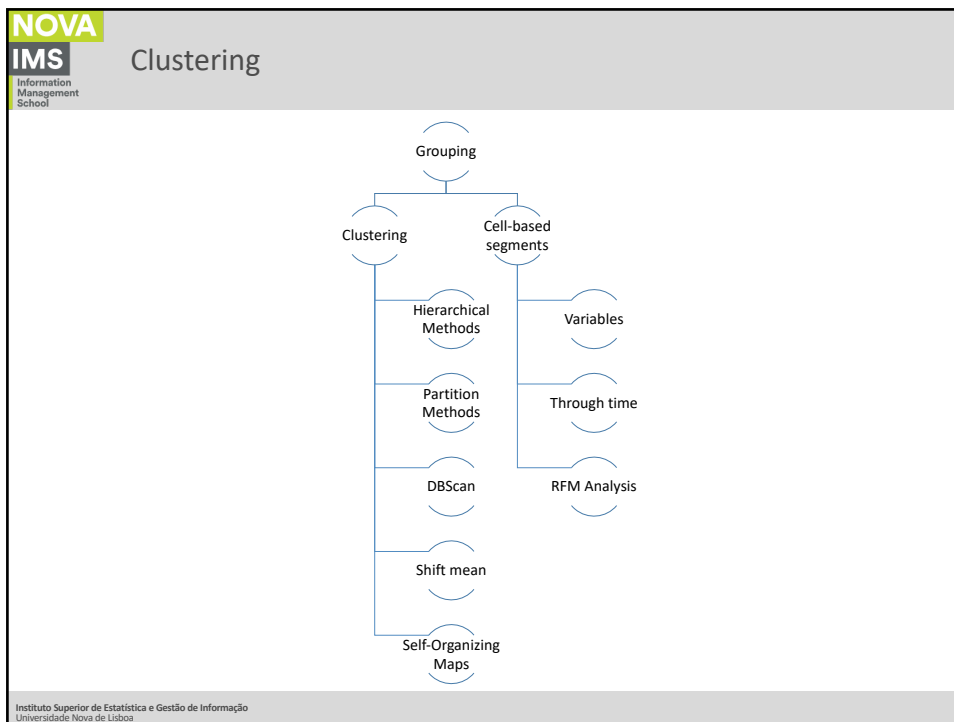
10



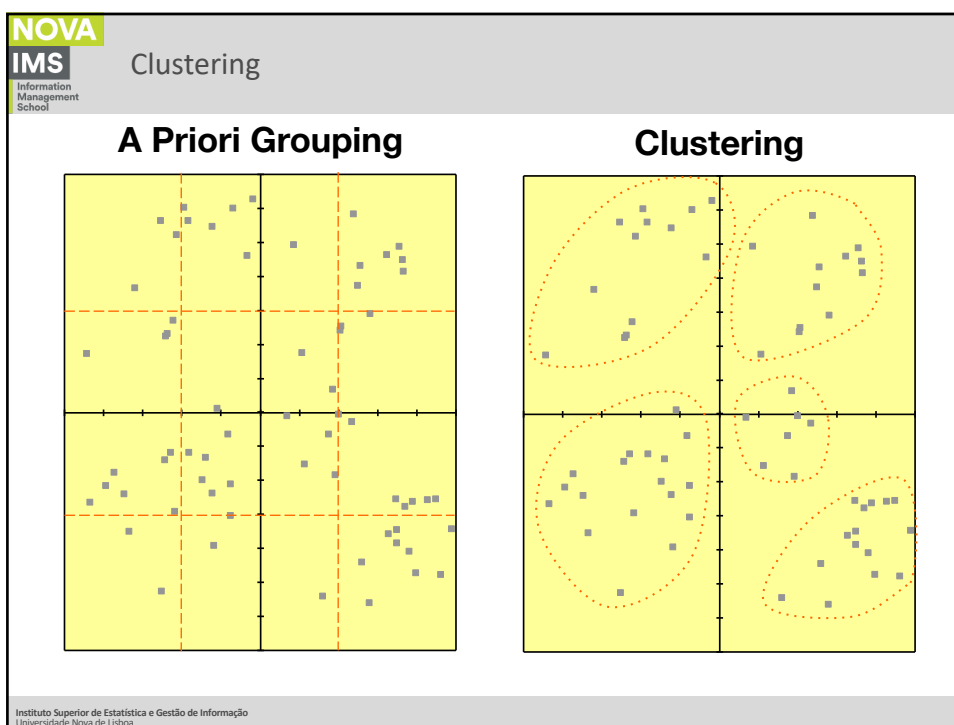
11



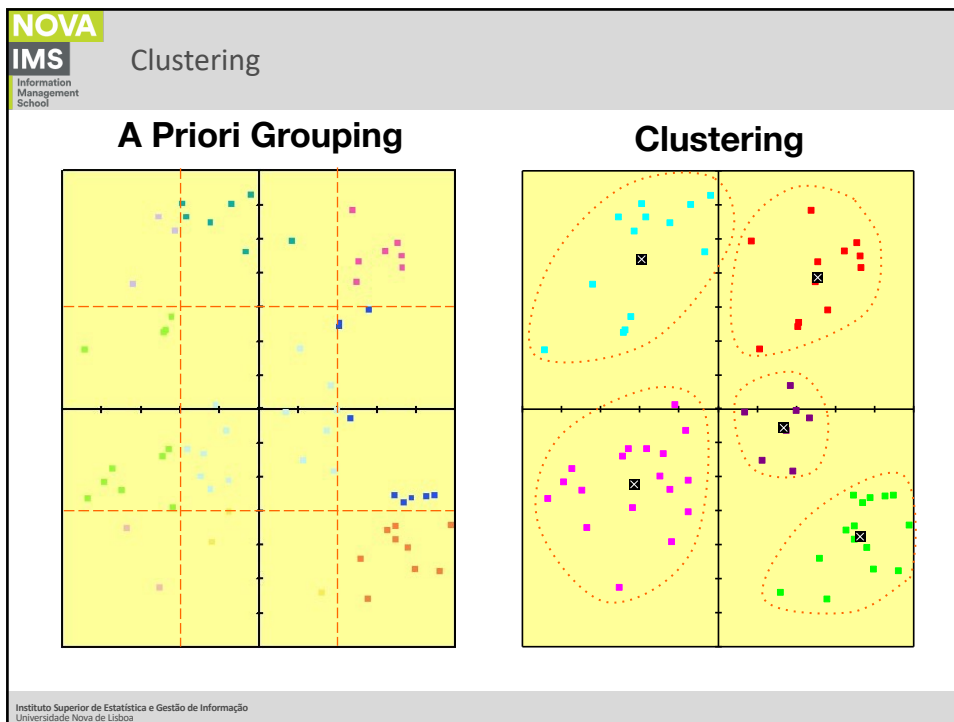
12



13



14



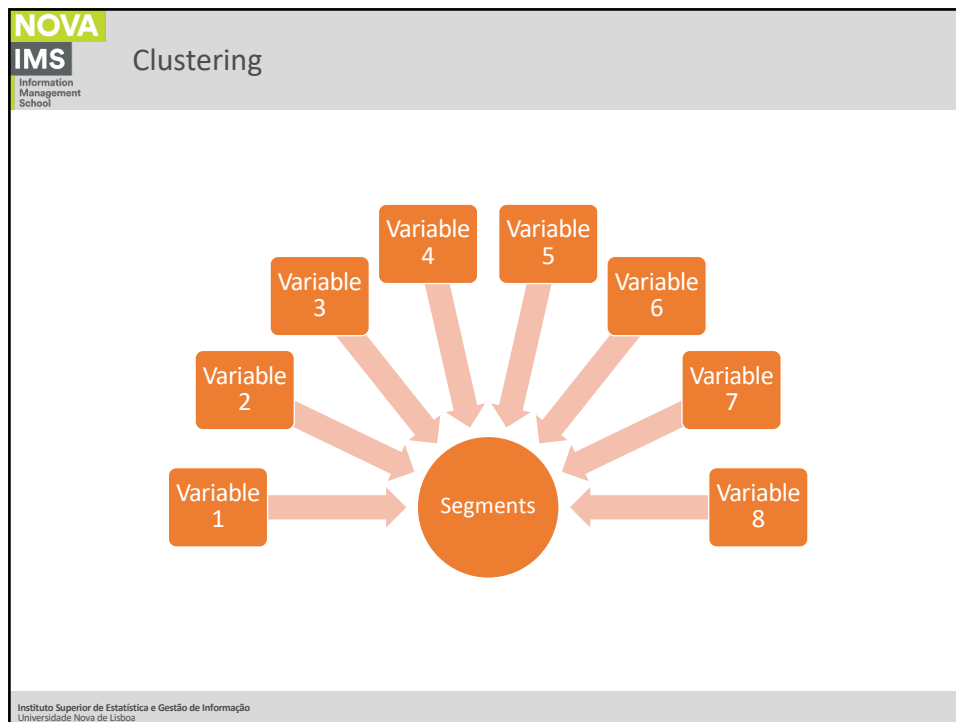
15

NOVA
IMS
Information
Management
School

Decide on the Variables to Use

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

16



17

NOVA
IMS
Information
Management
School

Clustering

- **Deciding which variables to use:**
 - **Objective of the segmentation**
 - Value/Engagement;
 - Needs;
 - Behaviors/Consumption;
 - GeoDemographics/Socio-economic characteristics.

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

18

NOVA
IMS
Information
Management
School

Clustering

- **Deciding which variables to use:**
 - The type of problem determines the variables to choose;
 - If the purpose is to group objects, the choice of variables with discrimination ability is crucial;
 - The quality of any cluster analysis is, first of all, conditioned by the variables used.

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

19

NOVA
IMS
Information
Management
School

Clustering

- **Deciding which variables to use:**
 - The choice of variables should replicate a theoretical context, a reasoning;
 - This process is carried out based on a set of variables that we know to be good discriminators for the problem at hand;
 - First of all, the quality of the cluster analysis reflects the discrimination ability of the variables we decided to use in our study.

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

20

NOVA
IMS
Information Management School

Similarity criterion

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

21

NOVA
IMS
Information Management School

Clustering

- **Similarity criterion:**
 - The analysis of similarity relations has been dominated by metrics based on Euclidean Spaces;
 - Objects as points in a multidimensional space, in a way that the observed dissimilarities between the objects correspond to distances between the respective points;
 - Thus, the use of clustering methods most times means the use of similarity ratios that respect these metrics:

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

22

NOVA
IMS
Information Management School

Clustering

- **Similarity criterion:**
 - In mathematics, a true measure of distance, called a *metric*, obeys three properties. These metric axioms are as follows, where d_{ab} denotes the distance between objects a and b :
 1. $d_{ab} = d_{ba}$ (measure is symmetric)
 2. $d_{ab} \geq 0$ and $= 0$ if and only if $a = b$ (distances are always positive except when the objects are identical)
 3. $d_{ab} \leq d_{ac} + d_{cb}$ (triangle inequality)

Exhibit 5.1 Illustration of the triangle inequality for distances in Euclidean space.

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

23

NOVA
IMS
Information Management School

Clustering

- **Similarity criterion:**
 - Euclidian distance: the distance between two elements (i, j) is the square root of the sum of the squares of the differences between i and j values for all variables ($v=1, 2, \dots, p$):

$$d_{ij} = \sqrt{\sum_{v=1}^p (X_{iv} - X_{jv})^2} \text{ euclidean also known as } L_2$$

$$d_{ij} = \sum_{v=1}^p |X_{iv} - X_{jv}| \text{ City Block or } L_1$$

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

24

NOVA
IMS
Information Management School

Clustering

- **Similarity criterion:**
 - Euclidian distance: the distance between two elements (i, j) is the square root of the sum of the squares of the differences between i and j values for all variables ($v=1, 2, \dots, p$):

$$L_1 : d_{ij}(1) = |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}|$$

$$L_2 : d_{ij}(2) = (|x_{i1} - x_{j1}|^2 + |x_{i2} - x_{j2}|^2)^{1/2}$$

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

25

NOVA
IMS
Information Management School

Clustering

- **Similarity criterion:**
 - Minkowski distance: is defined from the absolute distance, and can be considered as a generalization of both the Euclidean distance and the Manhattan distance. It coincides with Euclidean distance when $r=2$ and with Manhattan distance when $r=1$:

$$d_{ij} = \left(\sum_{v=1}^p |x_{iv} - x_{jv}|^r \right)^{1/r}$$

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

26

NOVA
IMS
Information
Management
School

Clustering

- **Similarity criterion:**
 - If a weight is assigned to each variable, according to their importance for the analysis, the weighted Euclidean distance takes the following form:

$$d_{ij} = \sqrt{\sum_{v=1}^p w_v (X_{iv} - X_{jv})^2}$$

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

27

NOVA
IMS
Information
Management
School

Clustering

- **Similarity criterion:**
 - Pearson correlation coefficient: its function is to measure the degree of linear correlation between two elements, for a number of variables:

Correlation

$$\rho(x_i, x_{i'}) = \frac{\sum_j (x_{ij} - \bar{x}_i)(x_{i'j} - \bar{x}_{i'})}{\sqrt{\sum_j (x_{ij} - \bar{x}_i)^2 \sum_j (x_{i'j} - \bar{x}_{i'})^2}}$$

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

28

NOVA
IMS
Information Management School

Choose the Algorithm

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa







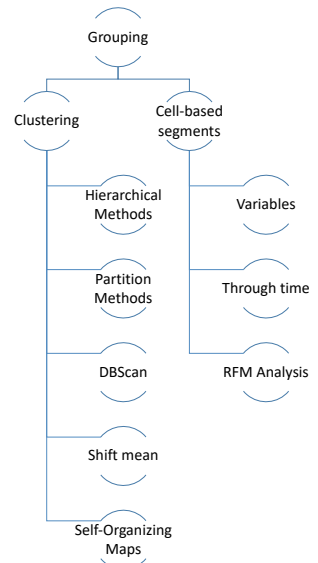





29

NOVA
IMS
Information Management School

Clustering



```

graph TD
    Grouping --> Clustering
    Grouping --> CellBased[Cell-based segments]
    Clustering --> Hierarchical[Hierarchical Methods]
    Clustering --> Partition[Partition Methods]
    Clustering --> DBScan[DBScan]
    Clustering --> ShiftMean[Shift mean]
    Clustering --> SelfOrganizing[Self-Organizing Maps]
    CellBased --> Variables[Variables]
    CellBased --> ThroughTime[Through time]
    CellBased --> RFM[RFM Analysis]
  
```

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

30

NOVA
IMS
Information Management School

A Priori Grouping

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

UNICIS

ACES

Schools

eduniversal

31

NOVA
IMS
Information Management School

Clustering

```

graph TD
    Grouping --> Clustering
    Grouping --> CellBasedSegments[Cell-based segments]
    Clustering --> HierarchicalMethods
    Clustering --> PartitionMethods
    Clustering --> DBScan
    Clustering --> ShiftMean[Shift mean]
    Clustering --> SelfOrganizingMaps[Self-Organizing Maps]
    CellBasedSegments --> Variables
    CellBasedSegments --> ThroughTime[Through time]
    CellBasedSegments --> RFMAnalysis[RFM Analysis]
  
```

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

32

NOVA
IMS
Information Management School

Cell-based segments

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

33

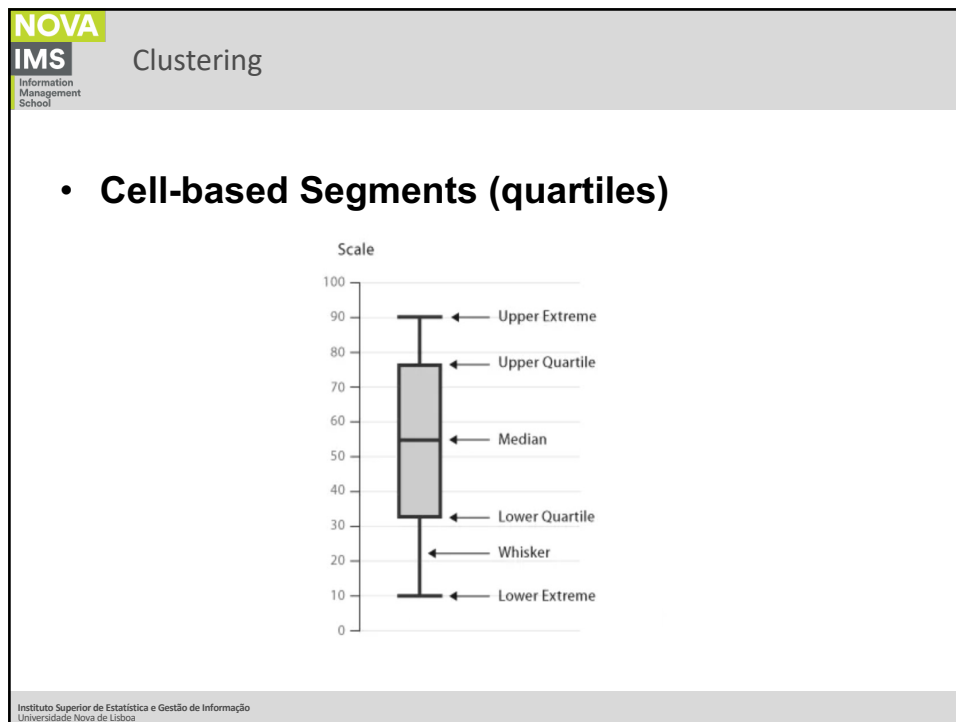
NOVA
IMS
Information Management School

Clustering

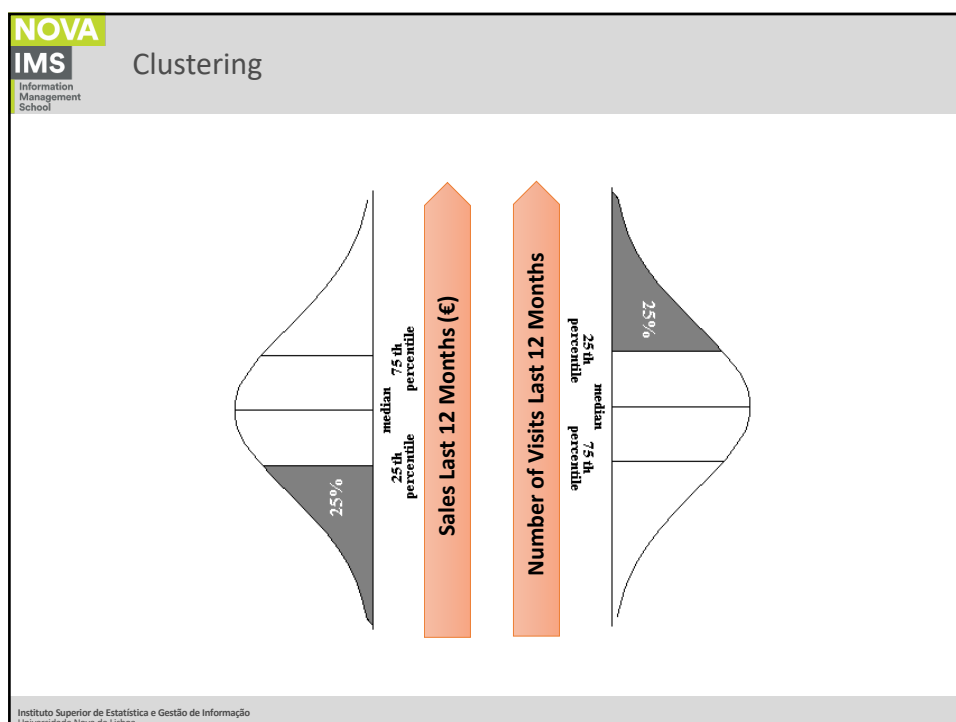
- **Cell-based Segments**
 - A percentile is a measure used in statistics indicating the value below which a given percentage of observations in a group of observations fall.
 - For example, the 20th percentile is the value (or score) below which 20 percent of the observations may be found.
 - The 25th percentile is also known as the first quartile (Q1), the 50th percentile as the median or second quartile (Q2), and the 75th percentile as the third quartile (Q3).
 - In general, percentiles and quartiles are specific types of quantiles.

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

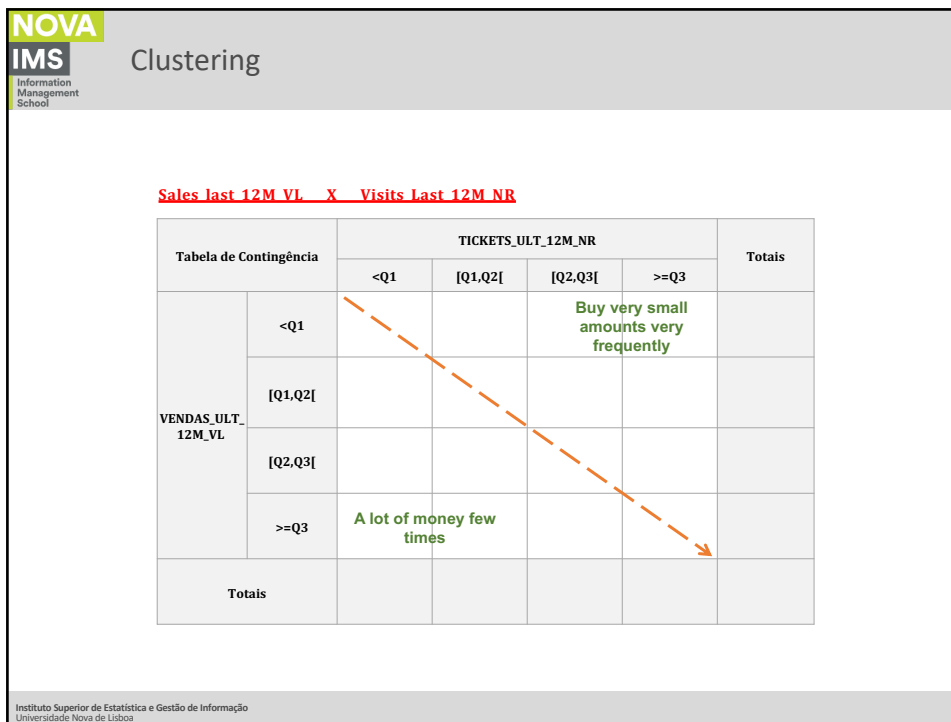
34



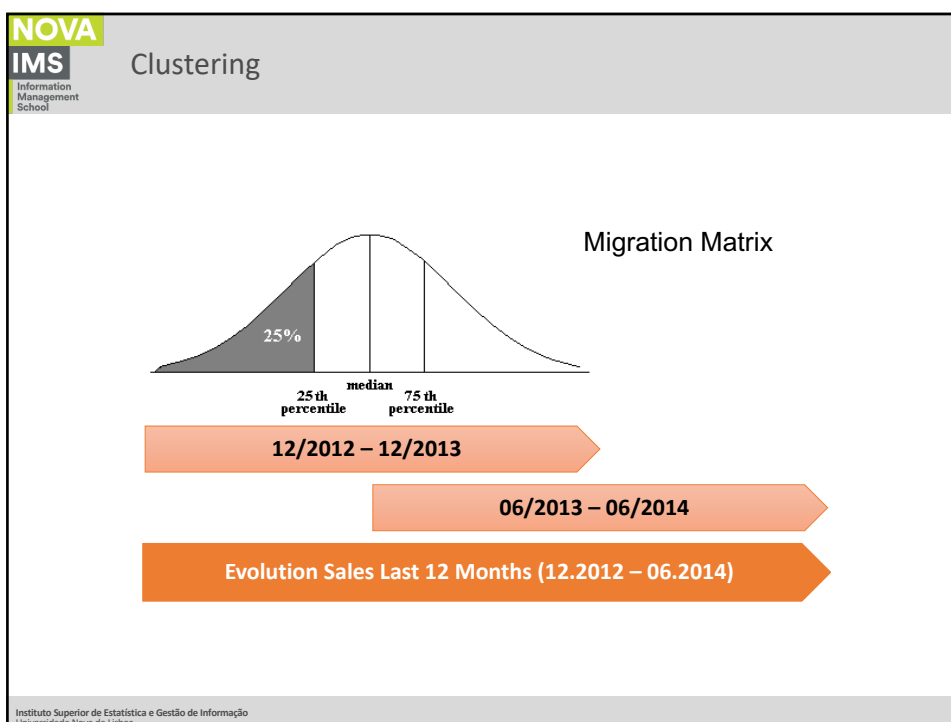
35



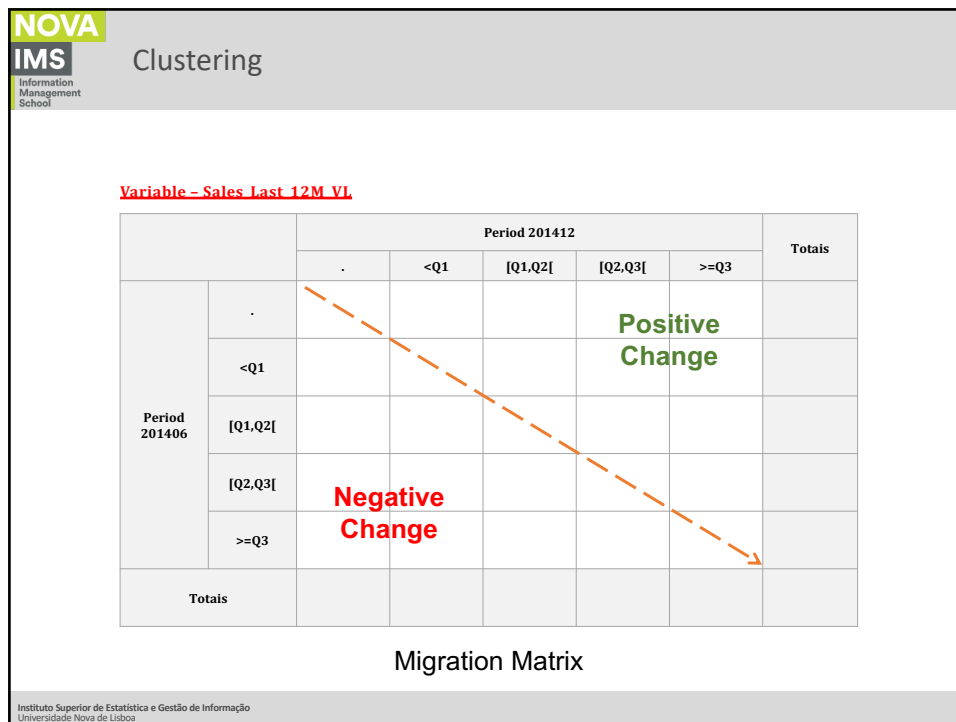
36



37



38



39

NOVA
IMS
Information Management School

Clustering

Tabela de Contingência		Period 06.2013/06.2014					Totais
		.	<Q1	[Q1,Q2[	[Q2,Q3[	>=Q3	
Periodo 12.2012/12.2013	.	0 a	30603 b	17050 c	8815 d	6427 e	62895 13.5%
	<Q1	28734 f	55411 g	13600 h	2772 i	178 j	100695 21.6%
	[Q1,Q2[	14834 k	15506 l	52421 m	16838 n	1097 o	100696 21.6%
	[Q2,Q3[	6540 p	3450 q	19346 r	59756 s	11608 t	100700 21.6%
	>=Q3	3765 u	760 v	1835 w	14733 x	79604 y	100697 21.6%
Totais		53873 11.6%	105730 22.7%	104252 22.4%	102914 22.1%	98914 21.2%	465683 100%

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

40

NOVA
IMS
Information
Management
School

RFM Analysis

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

UNICIS

ACES

Schools

eduniversal

41

NOVA
IMS
Information
Management
School

Clustering

- **RFM**
 - Based on the following principles:
 - Customers who have purchased more recently are more likely to purchase again;
 - Customers who have made more purchases are more likely to purchase again;
 - Customers who have made larger purchases are more likely to purchase again.

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

42

NOVA
IMS
Information
Management
School

Clustering

- **RFM**
 - Has been in active use in Direct Marketing for more than 40 years;
 - It can be used only for customer files that contain purchase history;
 - There are two methods:
 - Exact Quintiles;
 - Hard coding;

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

43

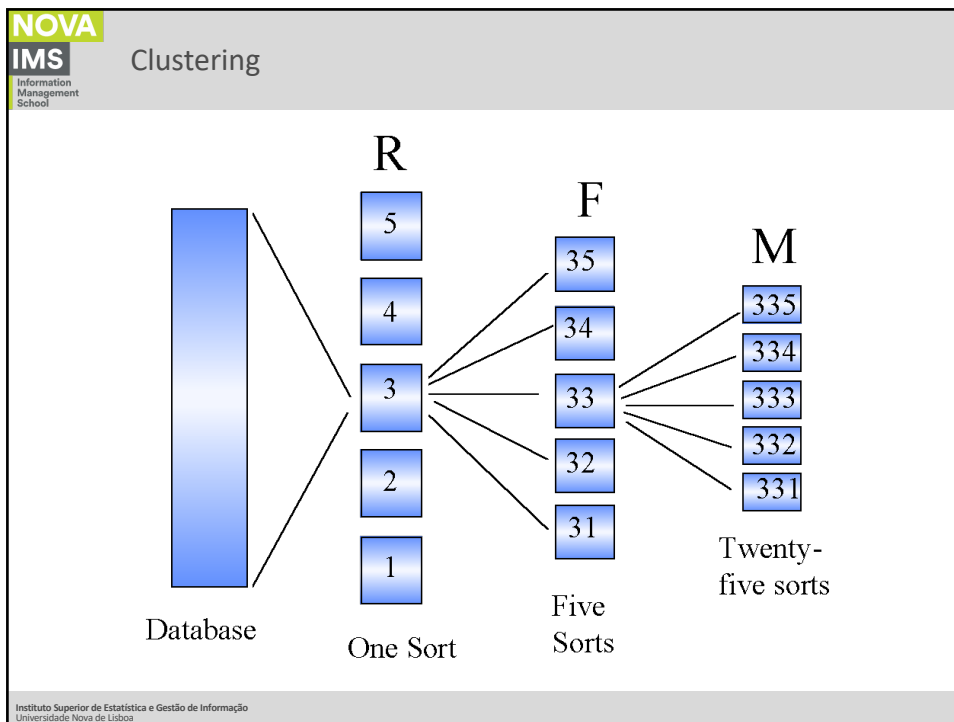
NOVA
IMS
Information
Management
School

Clustering

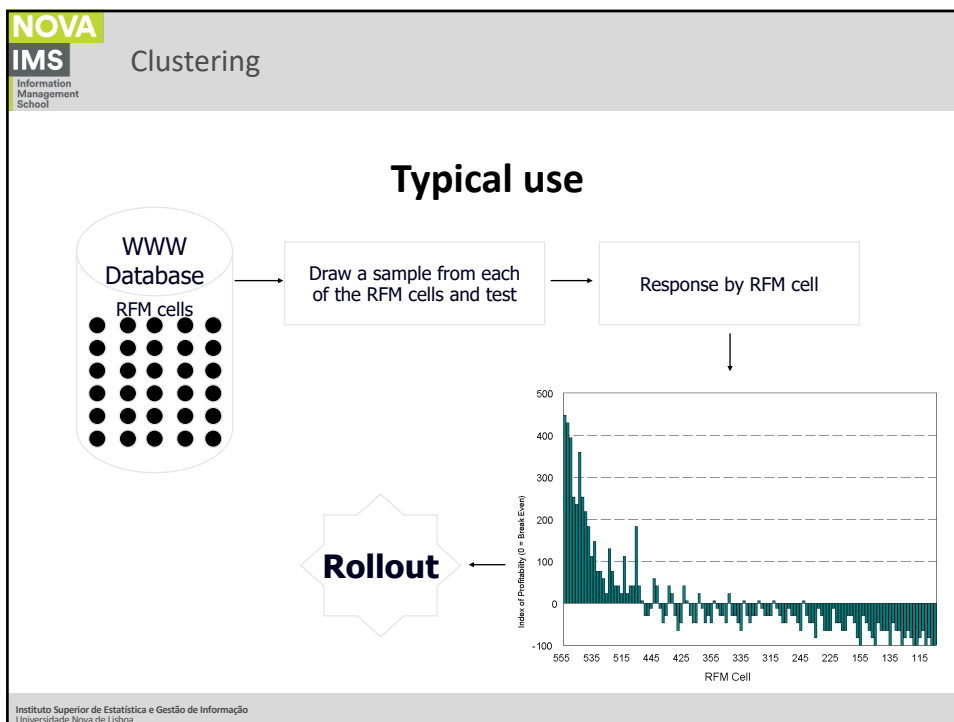
- **RFM**
 - How to do it (Exact Quintiles)?
 - We sort the database according to recency and divide into 5 quintiles (5 equal segments);
 - Do the same for the variables frequency and monetary;
 - Result: 125 cells of equal size (5*5*5).

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

44



45



46

NOVA
IMS
Information
Management
School

Clustering

- **RFM**
 - Its popularity comes from its simplicity, low cost and capacity to classify customers based on their behavior;
 - Opportunity to carry out tests in small, representative groups of each cell;
 - A more sophisticated modeling is almost always better, but is it worth it? Not always.

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

47

NOVA
IMS
Information
Management
School

Clustering

- **RFM**
 - Hard coding
 - Categories are divided by exact values (0-3 months; 4-6 months; 7-9 months; etc.);
 - More expensive in terms of programming, categories tend to change over time;
 - Very different quantities from cell to cell.

Instituto Superior de Estatística e Gestão de Informação
Universidade Nova de Lisboa

48

